



Conservation Laws:



Material Derivative:

Velocity Field:

PAC-NeRF: Physical Constraints

Conservation Laws:

$$\frac{D\sigma}{Dt} = 0 \quad \nearrow \text{Density Field} \quad \frac{D\mathbf{c}}{Dt} = 0 \quad \nearrow \text{Color Field}$$

Material Derivative:

$$\frac{D\varphi}{Dt} = \frac{\partial \varphi}{\partial t} + \mathbf{v} \cdot \nabla \varphi$$

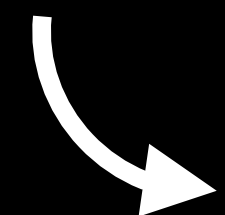
Velocity Field:

$$\rho \frac{D\mathbf{v}}{Dt} = \nabla \mathbf{T} \nearrow \text{Cauchy Stress Tensor} + \rho \mathbf{g}$$

PAC-NeRF: Governing Equations

$\mathbf{F}$ = Deformation gradient

$\mathbf{T}$ = Stress Tensor

 $J\mathbf{F} = \det(F)$

Material	Cauchy Stress Tensor
Elasticity	$J\mathbf{T}(\mathbf{F}) = \mu(\mathbf{F}\mathbf{F}^T) + (\lambda \log(J) - \mu)\mathbf{I}$
Newtonian Fluid	$J\mathbf{T}(\mathbf{F}) = \frac{1}{2}\mu \left(\nabla \mathbf{v} + \mathbf{v}^T \right) + \kappa \left(J - \frac{1}{J^6} \right)$
Plasticine	$J\mathbf{T}(\mathbf{F}) = \mathbf{U} \left(2\mu\epsilon + \lambda Tr(\epsilon) \right) \mathbf{U}^T$